

An investigation of nominal copular sentences in three reading paradigms: Acceptability judgments, self-paced reading, and eye-tracking

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Abstract

This work aims to investigate the elaboration of (nominal) copular sentences in three different experimental paradigms involving a reading task: an acceptability judgment, a self-paced reading and an eye-tracking experiment. Nominal copular sentences (NCs), such as [DP1 The picture of the wall] is [DP2 the cause of the riot], represent a challenging phenomenon for, at least, two reasons: (i) they can be distinguished in two subtypes, namely canonical and inverse NCs, related to the different order of the DPs (respectively, [DP_{subject} is DP_{predicate}] in canonical form vs. [DP_{predicate} is DP_{subject}] in inverse form); (ii) these two subtypes are associated with one and the same type of string [DP is DP], although their underlying structure is completely different. Our results show that no differences emerge in the off-line paradigm, i.e. in the acceptability judgments. On the other hand, the self-paced-reading task and the eye-tracking experiments show an asymmetry between these two types of NCs, with higher processing costs for inverse NCs. More specifically, the DP_{subject} is looked at more often and for longer times in inverse NCs. Moreover, when comparing the DP_{subject} to the DP_{predicate} in postverbal position in the eye-tracking experiment, sentence structure resulted as a good predictor for total reading time and regression path duration, even after taking into account the length and frequency of the words used. These results strongly support the hypothesis that syntactic structure is a primary factor in generating a different reading pattern between the same string of lexical types of items.

Keywords: Syntax; Psycholinguistics; Sentence processing; Reading; Nominal copular sentences; Eye-tracking

SVO Languages like Italian and English usually display the order DP_{subject}-verb-DP_{object} showing some asymmetries between the preverbal DP and the post verbal DP, regardless of the verb that is involved. Consider, among many other asymmetries, the fact that the DP_{subject} usually triggers the agreement on the verb, while the DP_{object} does not ([John]_i+- sing is+sing eating [many fruits]_j+plu), or, again, the fact that the DP_{subject} displays restrictions w.r.t. movement transformation that the postverbal DP does not display (e.g., that-trace effect in English, the lack of cliticization in Italian, etc.). These asymmetries, often referred to as “subject–object” asymmetries, provide significant empirical support for the assumption of an asymmetric representation of sentences in natural language syntax (Chomsky, 1986, 62), as shown by the following schema:

(1) [DP_{subject} [VP [V DP_{object}]]]

Moro (1991, 1997) successfully showed that when V is realized by the copula, this representation fails, as witnessed by many empirical tests in several languages, such as in case of extraction. To see such asymmetries, first consider two subtypes of nominal copular sentences (NCs): the canonical NC (2a), in which the order of the grammatical functions is subject-copula-predicate, and the inverse NC (2b), in which the order is inverted, i.e., predicate-copula-subject:

- (2) a. [DP_{Subj} A picture of the wall] is [DP_{Pred} the cause of the riot]
 b. [DP_{Pred} The cause of the riot] is [DP_{Subj} a picture of the wall]

Given the same DP-is-DP order, no differences should be expected in these two sentences, but this is not the case. Consider, for example, the (wh-sub-)extraction – both in Italian and English – from either the DP_{Subj} or the DP_{Pred} in both canonical and inverse NCs, as originally discussed in Moro (1997) and experimentally investigated in Greco et al. (2020): (i) no sub-extraction is allowed from the DP_{Subj} in both canonical (3a) and inverse NCs (3b), whereas (ii) it is allowed from the DP_{pred.} in canonical NCs (3c), namely when it occurs in the postverbal position, while it is not allowed in inverse NCs (3d), namely when DP_{pred} occurs in the preverbal position:

- | | | | | | | |
|-----------------|-----------|----------------------|-------------------|--|-----|--|
| (3) a. | *di quale | muro _i | pensi che | [DP _{Subj} una foto t _j] | sia | [DP _{Pred} la causa della rivolta] _i ? |
| a ^{it} | *which | wall _i | do you think that | [DP _{Subj} a picture of t _j] | was | [DP _{Pred} the cause of the riot] _i ? |
| b. | *di quale | muro _i | pensi che | [DP _{Pred} la causa della rivolta] _i | sia | [DP _{Subj} una foto t _j] _i ? |
| b ^{it} | *which | wall _i | do you think | [DP _{Pred} the cause of the riot] _i | was | [DP _{Subj} a picture of t _j] _i ? |
| c. | di quale | rivolta _i | pensi che | [DP _{Subj} una foto del muro] _i | sia | [DP _{Pred} la causa t _j] _i ? |
| c ^{it} | which | riot _i | do you think | [DP _{Subj} a picture of the wall] _i | was | [DP _{Pred} the cause of t _j] _i ? |
| d. | *di quale | rivolta _i | pensi che | [DP _{Pred} la causa t _j] _i | sia | [DP _{Subj} una foto del muro] _i ? |
| d ^{it} | *which | riot _i | do you think that | [DP _{Pred} the cause of t _j] _i | was | Subj a picture of the wall] _i ? |

Such asymmetries between canonical and inverse NCs are widespread across languages. English and Italian have already been illustrated above, representing examples from the Germanic and Romance language families, respectively. Other languages have also been examined (e.g., Bartošová 2017; Arche, Fábregas & Marín 2019), revealing a similar pattern, albeit with some language-specific variations. For example, the agreement pattern between the copula and the subject or predicate DP in Italian and English is different (see example 7 below). An analysis of NCs as raising of a single DP from an underlying “small clause” (SC) structure – i.e., the structure that results from merging two DPs of the form [SC DP DP] (Chomsky, 2013; Moro, 2000; Rizzi, 2016) – takes into account such asymmetries. More specifically, Moro (1988, 1997), building on the Stowell’s (1978) idea of the verb to be as a raising verb, suggests that either the subject DP or the predicate DP can raise to the pre-verbal position from the SC, driven by the instability of symmetric structures (Moro, 2000): if the subject DP raises to the pre-verbal position while the predicate DP remains in situ (5a), this results in a canonical NC sentence; conversely, if the predicate DP raises, while the subject DP stays in situ, this results in an inverse NC sentence. The following representation summarizes the

different syntactic structures:

- | | | | | |
|-----|----|---|-----------|-----|
| (4) | a. | [_{TP} DPsubj [_{VP} V°copula [_{SC} t DPpred]]] | Canonical | NCs |
| | | A picture of the wall is the cause of the riot. | | |
| | b. | [_{TP} DPpred [_{VP} V°copula [_{SC} DPsubj t]]] | Inverse | NCs |
| | | The cause of the riot is a picture of the wall. | | |

Crucially, representations in (4) yield one and the same type of string [DP is DP], regardless of the two NC subtypes, but with different syntactic representations. Roughly speaking, in both structures the subject position is filled by a NP, but of a different kind, either the subject or the predicate NP. This allows to derive all the contrasts in (3), and many others, on the basis of a principled framework since different subadjacency conditions are associated to different syntactic positions (see [Chomsky, 1986](#); [Cinque, 1990](#)).⁴ Clearly, only in the case of canonical copular sentences is the standard order found, where the subject precedes the verb and the predicate follows. In inverse copular sentences, however, the predicate moves up and therefore appears in a pre-verbal position that is normally reserved for subjects. It's worth noting that the informational force associated with the different order of DPs alone does not fully explain the asymmetries between canonical and inverse copular sentences. More specifically, consider, for example, the subject of inverse copular sentences that is inherently endowed with focal interpretation, as witnessed by the following minimal pair:

- (5) a. John was [t the culprit]. (canonical)
b. The culprit was [John t]. (inverse)

These two sentences, although *prima facie* may seem completely synonymous, do convey different information. This is clear if we negate them and construe contrasts like:

- (6) a. John was not the culprit but the witness
b*. John was not the culprit but Peter
c. The culprit was not John but Peter
d*. The culprit was not John but the witness

This simple paradigm, which sharply contrasts with any other verb than the copula, indeed shows that the formal difference attributed structurally to raising robustly correlates with an independent asymmetry in terms of informational force. Indeed, however, without assuming a raising process to preverbal position (again, of either DP, namely the subject and the predicate yielding respectively canonical vs. inverse copular sentences), these contrasts stemming from the very same sequence of syntactic categories, i.e. DP copula DP, would just constitute a complete mystery.

1.1. Experimental works on copular sentences

Further evidence on the different contribution of *be* in copular constructions and in auxiliary constructions come from several experimental studies (see, among many others, [Hegde, 1980](#); [Aaronson and Ferres 1984](#)). For example, [Samar and Berent \(1991\)](#) collected psycholinguistic evidence in favor of the raising analysis of copular sentences, but not of the nominal type. More specifically, they tested copular sentences with adjectives (e.g., 'College students [VP are [AdjP friendly]]') to assess the presence of a subject trace, which would be expected according to a raising analysis of copular structures (e.g. [_{TP} DPsubj [_{VP} V°copula [_{SC} t AP]]]). Their results support a raising analysis of copular sentences by providing experimental evidence in favour of the presence of a trace in the original subject position. However, copular sentences with adjectives cannot show the alternation between canonical and inverse sequence (e.g., '*[AdjP friendly] [VP are) college students'), which is the main research interest of the present work. Canonical vs inverse NCs

presenting the structure exemplified in (4a-b) have been investigated by [Greco et al. \(2020\)](#) in a series of acceptability judgement tasks in which sentences were presented along with a seven-point acceptability scale. They found that adult native speakers of Italian consistently rated inverse copular sentences as less acceptable than their canonical counterparts. However, a possible morphosyntactic cue occurred in their stimuli: since Italian is a prodrop language, it displays a copula that consistently agrees with the subject DP, whether it appears in the pre-verbal or post-verbal position (7a-b). This contrasts with languages like English, which enforce agreement with the preverbal DP, regardless of its predicative or argumental status:

- (7) a. [DP_Subject Le foto del muro] sono / *è [DP_Predicate la causa della rivolta]
 The pictures of the wall are / *is the cause of the riot
 'The pictures of the wall are the cause of the riot'
 b. [DP_Predicate La causa della rivolta] sono / *è [DP_Subject le foto del muro]
 the cause of the riot are / *is the pictures of the wall
 'The cause of the riot is the pictures of the wall'

The Italian agreement pattern, therefore, constitutes an overt cue to the role assignment in copular sentences with a plural subject, leading to a possible number mismatch effect in those types of experimental designs.

1.2. Sentence processing

From a psycholinguistic perspective, by postulating different subadjacency conditions between the two NPs, this theory also predicts different processing costs associated with canonical vs. inverse copular sentences. However, to the best of our knowledge, the online processing of (nominal) copular sentences has never been addressed. As we have seen, this type of sentence presents the exact same surface structure (DP-is-DP) with the only difference being the different function (subject vs. predicate) of the two NPs. Following the “raising” approach, the relevant differences between canonical and inverse NCs are purely syntactic, thus making this type of structure an excellent test of how syntax can influence sentence processing.

While several human parsing mechanisms have been discussed since the 1960s, two models have often been contrasted: models that rely on “heuristic” strategies, where sentence comprehension follows distributional regularities, such as the DP-verb-DP sequence (with the agent preceding the patient), as in transitive declarative sentences ([Ferreira, 2003](#); [Bever, 1970](#); [Fodor et al., 1974](#)); and those that employ an “algorithmic” approach to sentence comprehension ([Frazier, 1987](#); [Frazier and Clifton, 1996](#); [Gibson, 2000](#); [MacDonald, 1994](#)), where an algorithmic process incrementally constructs a detailed syntactic representation that captures both the phrasal structure of the sentence and the grammatical dependencies between phrases (see [Bader and Meng, 2018](#) for a detailed discussion). The tension between these models has led to many intermediate proposals, such as the so-called “good-enough” model (see [Christianson et al., 2001](#); [Ferreira, 2003](#), and subsequent works), where both algorithmic and heuristic routes operate independently and in parallel. Since the heuristic processes operate faster, however, their interim output strongly influences the algorithmic parser ([Karimi and Ferreira, 2016](#)), leading to misinterpretation effects and lingering effects in garden-path sentences ([Christianson et al., 2001](#)).

The good-enough model and purely heuristic models have been challenged (see [McLean et al., 2022](#) and the references therein), as they fail to provide accurate predictions, particularly in online experiments. This includes studies of passive vs. active constructions in both self-paced reading ([Paolazzi et al., 2019](#); [Cutter et al., 2022](#)) and eye-tracking experiments ([Paolazzi et al., 2022](#)). These latter findings are especially important because they highlight a difference between offline and online experimental paradigms: passives were processed more quickly than actives (online data) – contrary to the heuristic model – but led to more comprehension errors (offline data).

Inverse nominal copular sentences represent a distinctive type of non-canonical sentences in that, unlike passives and clefts, their surface form is identical to their canonical counterparts. Although the sequence of elements displays the same distributional regularity – the DP-is-DP order in both

canonical and inverse NCs corresponds to the typical structure of transitive declarative sentences (Bever, 1970) – their differing functions are determined at a structural level, where grammatical dependencies are established. This ultimately calls for an algorithmic approach to sentence comprehension: the type of “raised” element in NCs (i.e., either the subject or the predicate) is predicted to bypass heuristic strategies, and with regard to the good-enough model, although both heuristic and algorithmic routes are activated, the absence of a specific syntactic probe (e.g. a comprehension question) targeting the NPs roles within the task would lead to syntactically underspecified representations.

Although theoretical accounts do not always allow for straightforward computational predictions, the psycholinguistic investigation of NCs could help shed light on the mechanisms of the parser. Since the inverse NCs could only be detected algorithmically, any additional processing cost could be ascribed not only to syntactic complexity leading to cognitive effort, but also to the specific architecture of sentence processing.

1.3. The present study

The present study aims to explore the processing of NCs, which in their inverse form constitute a more complex syntactic construction (see above), with three different methodologies: an acceptability judgment, a self-paced reading and an eye-tracking paradigm. Therefore, the main goal of the research is to verify whether the syntactic complexity postulated by Moro's (1997, 2017) raising analysis translates into a greater cognitive effort in processing. This will be achieved through different experimental paradigms to provide a fine-grained description of the difficulties encountered while reading such sentences. Finally, this in-depth analysis will be discussed in light of the current issues in sentence processing, given the evident contrast between linear order and deep asymmetries within NCs.

We expect different outcomes from online and offline methodologies, following previous research stating that the interpretation of noncanonical sentences can depend on the experimental paradigm (Paolazzi et al., 2019) and on the demanded task (Bader and Meng, 2018). In the specific case of NCs, based on Greco et al. (2020), we expect the difference between canonical and inverse NCs to affect acceptability judgments (offline data), with inverse NCs being less acceptable than canonical NCs (see the discussion in the introduction). The stimuli presented in our study were adapted from Greco et al.'s (2020) paradigm, however, we maintained both noun phrases in the singular form to examine whether the observed effect was attributable to the plural agreement of the copula. If this were the case, no acceptability differences would be expected between canonical and inverse structures.

As for online processing, heuristic and good-enough models predict that no difference should appear in the computation of either form on NCs. On the other hand, according to the algorithmic model, differences should emerge, given that the syntactic dependency in inverse NCs where the predicate raises to the subject position is believed to entail higher cognitive effort on the parser (Greco et al., 2023). Therefore, we expect the real-time processing of inverse NCs to be more demanding compared to canonical NCs (online data). More specifically, in a self-paced reading task, this would result in higher reaction times in the inverse copular sentences, arguably on both DPs and on the final region. However, since reaction times are more likely to reflect immediate processing difficulty (Boland and Blodgett, 2002; Frazier and Rayner, 1982; McElree et al., 2001), an eye-tracking experiment was run to investigate late measures as well, that are traditionally linked to the processing of the syntactic structure of the sentence. Since early measures are influenced mostly by lexical properties, we do not expect first fixation and gaze duration to be modulated by the NCs form (canonical vs inverse), regardless of the parsing mechanism involved. With respect to late measures, particularly those associated with reanalysis such as total reading times and regression path duration, differences are expected only in the case the difference in the deep structure between conditions is detected algorithmically.

All three experiments presented in this paper were approved by Ethics Committee of the Department of Brain and Behavioural Sciences of the University of Pavia.

2. EXPERIMENT 1

The first experiment was an acceptability judgement task, meant to assess whether the asymmetry between canonical and inverse NCs is detected by participants in terms of naturalness and comprehensibility of the sentence. This first task is a replication of the one proposed in [Greco et al. \(2020\)](#), with one crucial modification: the copula was kept singular, in order to avoid any number mismatch effect due to the agreement to the postverbal subject in the inverse condition.

2.1. Methods

The acceptability judgement task was implemented in PennController for Internet Based Experiments (PCIBex, [Zehr and Schwarz, 2018](#)). Eighty adult native speakers of Italian were recruited online through social media. Participants with ages above 45 were excluded in order to keep the age range uniform across studies, and therefore 69 participants were included in the analyses (age range: 18–45, mean age: 28; 51 females, 16 males, 2 who preferred not to say). After providing informed consent, participants were asked to read each sentence and then rate its naturalness and comprehensibility on a 7-point Likert scale, where 1 was defined as ‘not at all’ acceptable and 7 as ‘entirely’ acceptable. The experiment began with four practice trials, then the trials within the experimental phase were presented in two blocks. Counterbalanced lists were created and randomly assigned to participants to ensure that each item was presented in the two experimental conditions, but not to the same participant.

2.2. Materials

The experimental materials consisted of the same stimuli used in [Greco et al. \(2020\)](#). However, those sentences presented a possible confounding factor in the copula plural agreement, which makes it an overt cue to the role assignment of incoming material (see above). Therefore, we decided to take away the possible confounding variable in the copula agreement by using singular DPs for both the subject and the predicate DPs. In order to take away any other possible confounding variable, we also employed experimental items showing the same underlying structure of the prototypical copular sentence in (6), i.e., ‘La foto del muro è la causa della rivolta’ (engl. ‘the picture of the wall is the cause of the riot’). All the DPs displayed the form [DP1 [PP of [DP2]]] and all the lexical noun heads were inanimate, definite, and countable. Noun heads, either the subject or the predicate DP (e.g., foto [picture] / causa [cause]), were assigned both external and internal h-roles. In the case of picture, the external argument was the agent of the action of taking the picture, while the internal object was the theme of the action, i.e. the embedded PP (e.g. del muro, Eng. ‘of the wall’). Similarly, the lexical noun cause was assigned both an external (agent) h-role and an internal (theme) h-role, the latter being the embedded PP (e.g. della rivolta, Eng. ‘of the riot’). Therefore, the internal h-role of both the DP_{subject} and the DP_{predicate} was represented by the embedded PP, which was specifically linked to its selecting DP and could not be swapped. Accordingly, all DPs were symmetric from a structural point of view. Crucially, an asymmetric relationship between the two DPs was introduced to discriminate the subject vs. predicate function: the DP_{subject} (e.g. the picture) represents the external argument of the DP_{predicate} (e.g. the cause), but not vice versa, since cause can select picture but picture cannot select cause. Given that all DPs were in the singular form (contra [Greco et al., 2020](#)), no morphological clues such as the agreement were available in discriminating the role of the two DPs. Therefore, the difference between subject and predicate function was established on a pure syntactic ground.

The critical sentences were arranged in duplets, with two levels of manipulation (copular sentence type: canonical or inverse). 8 critical duplets (16 items) were spaced out by 94 fillers of varying degrees of acceptability (low, medium, high). The critical items were the same as the [Greco et al.'s \(2020\)](#) second experiment, with two main differences: the copula was kept singular, as described above, and in addition the sentence was turned into a subordinate clause, and appeared after a main clause with the structure “X said that”. The aim of this second modification was to avoid the occurrence of a critical part of the sentence (i.e. the predicate or subject noun phrase) being the first element of the sentence to be read. The modification also made the copular sentence sound more natural and less abrupt by giving it more context. An example of experimental item in the two critical conditions is the following:

- (8) a. *Il professore ha detto che la foto del muro è la causa della rivolta.*
 The professor say.3rd.sing.past that the picture of.the wall be.3rd.sing.pres the cause of.the
 'The professor said that the picture of the wall is the cause of the riot'
- b. *Il professore ha detto che la causa della rivolta è la foto del muro.*
 The professor say.3rd.sing.past that the cause of.the riot be.3rd.sing.pres the picture of.the wall.
 'The professor said that the cause of the riot is the picture of the wall'

2.3. Results

All participants correctly rated filler sentences as scarcely, averagely, or highly acceptable, and thus no participants were excluded from data analysis.

As can be appreciated by looking at [Table 1](#), the experimental sentences were considered acceptable by participants. Data analysis on the acceptability ratings was performed with Cumulative Link Mixed Model with a logit link function, implemented via the `clmm()` function from the `ordinal` package in R ([Christensen, 2023](#)). Condition was the only fixed factor in the analysis, while random intercepts were included for participants and items, and a by-participant random slope for condition was added. This model was selected based on model comparison using the `compare_`

Table 1
Descriptive statistics for the results of Experiment 1.

Condition	Mean acceptability	Standard error	Median acceptability
Canonical	5.93	0.06	6
Inverse	5.83	0.07	6

`performance()` function from the `easystats` package ([Lüdtke et al., 2021](#)), as it was the one with the lowest AIC score. The results indicated that there was no significant difference between how natural and comprehensible the two types of copular sentence were perceived to be ($b = 0.11$, $SE = 0.07$, $z = 1.56$, $p = 0.12$).

By looking at the results, however, it came to our attention that two experimental items had acceptability ratings that were lower than the others, showing a median below 6. These items were deemed non optimal and excluded from following experiments.

2.4. Discussion

The results obtained in the acceptability judgement task paint a different picture compared to the previous experiment presented in [Greco et al. \(2020\)](#), in which the same stimuli were employed. Although the acceptability ratings for the canonical sentence are similar in the two experiments, inverse NCs were rated lower in the previous study, and the difference between canonical and inverse copular sentences was indeed significant, while it is not in the present study. This incongruence between the two studies can be attributed to the fact that, in the present design, the number mismatch between subject and predicate was eliminated. As previously argued, Italian

native speakers generally have a preference for the subject being presented in pre-verbal condition (as also confirmed in [Greco et al.'s \(2020\)](#) third experiment). The number mismatch present in inverse sentences between the pre-verbal predicate and the copula makes it more evident to a naive speaker that the subject is indeed not in its preferred position.

In the present study, we aimed to eliminate the potential confounding factor of the number mismatch, in order to make sure that the difference between the two conditions was driven only by the syntactic asymmetry, not cued by any morphosyntactic features such as agreement. Based on the results obtained, syntactic derivation per se does not seem to interfere with sentence comprehensibility and naturalness, as consciously judged by native speakers. This result, however, does not speak as to whether syntactic complexity may induce additional processing costs during parsing, which constitutes our main research aim. Assessing the processing cost of syntactic complexity was beyond the scope of experiment 1. Instead, this first experiment served to ensure that the experimental stimuli were optimally designed for our specific experimental aim, allowing us to test syntactically noncanonical sentences without the addition of any other layer that might affect processing. In order to further explore the weight of syntactic complexity, we investigated the same items in a further experiment through a self-paced reading paradigm.

3. EXPERIMENT 2

As an online methodology, SPR has the advantage of capturing cognitive processing in real time. Sentences are presented in segments, one at the time, and participants can press a button to reveal the next segment, causing the previous one to disappear. This allows participants to control their reading speed, making their comprehension of the sentence self-paced. The time taken to read each segment and press the button (i.e., reaction times or RTs) is viewed as an indicator of cognitive effort ([Jegerski, 2013](#)). If the syntactic derivation of inverse copular sentences entails higher processing cost, we expect to see inflated reading times in this condition as opposed to canonical sentences.

3.1. Methods

The self-paced reading task was implemented in PennController for Internet Based Experiments (PCIBex, [Zehr and Schwarz, 2018](#)) as well. Eighty adult native speakers of Italian (age range: 21–45; mean age: 30; 43 females, 37 males) were recruited online through the platform Prolific. Prior to the experiment, all participants were asked to report any learning and/or language disorder and to agree to the informed consent. Each participant was rewarded 1.90€ for his/her/their participation. Except for the two items that were not judged as highly acceptable in Experiment 1, the same sentences were presented in a masked chunk-by-chunk non-cumulative self-paced reading paradigm, counterbalanced in lists as in the previous experiment. Chunks consisted of up to three words, and were not necessarily corresponding to linguistic phrases. An example of how the sentences in (8) were presented is given in [Fig. 1](#).

Il professore						
	ha detto che					
		la foto				
			del muro			
				è		
					la causa	
						della rivolta.

Fig. 1. Example of the presentation of the stimuli in Experiment 2, the sentence presented can be translated as The professor | said that | the picture | of the wall | is | the cause | of the riot, as seen in example (8a).

All critical items followed the same structure and were always composed of 7 chunks:

Chunk 1: a noun phrase composed of definite article and common noun;

Chunk 2: the verb “to stay” in perfective tense and the conjunction ‘che’;

Chunk 3: a noun phrase composed of a definite article and a common noun (which can serve as subject or predicate in the NC structure);

Chunk 4: a prepositional phrase composed of the preposition “of” + a definite article and a common noun;

Chunk 5: the copula “è”;

Chunk 6: a noun phrase composed of a definite article and a common noun (which can serve as subject or predicate in the NC structure);

Chunk 7: a prepositional phrase composed of the preposition “of” + a definite article and a common noun;

In this task as well, participants first completed 4 practice trials, before starting an experimental session that was divided in two blocks. In order to ensure reading for comprehension, every item was followed by a question, with an equal number of “yes” and “no” answers (e.g., “Is the cause of the riot a picture?” following the sentence depicted in Fig. 1). Feedback after each question was provided only in the practice trials. As for the comprehension question during the experimental session, it was established that participants who answered more than 25 % of the questions following filler items incorrectly would be excluded from data analysis, under the assumption that they were not paying attention during reading. The task lasted 16 min on average.

3.2. Data analysis and results

No participant was excluded from data analysis due to low accuracy on the comprehension questions after filler items. The reaction times (RTs) for Chunks 3 to 7 were analyzed, as to include the whole NC structure. Only RTs from sentences where participants correctly answered the comprehension questions were included in the statistical analysis. Since overall accuracy was 96 % in both conditions, this exclusion had no substantial impact on the data. For data trimming, RTs below 100 ms or above 3000 ms were removed. Then a logarithmic transformation was performed to normalise the RT data distribution (following Baayen, 2008).

For each chunk, a linear mixed model (LMM) was fitted for data analysis, using the `lmer()` function from the `lme4` package in R (Bates et al., 2015) with log-transformed RTs as a dependent variable and the type of NC sentence as a fixed factor. For all Chunks excluding Chunk 5, which only included the copula, the length and frequency of the noun presented within the Chunk were also included as fixed factors in the model. The random effects structure for each model was determined using a data-driven approach, systematically performing model comparisons using the `compare_performance()` function from the `easystats` package (Lüdtke et al., 2021) and choosing the model with the lowest AIC score. The mean RTs in each chunk can be seen in Fig. 2. The results of the statistical analysis of RTs show that, as can be expected, length was a good predictor for RTs at Chunks 3, 4 and 6. However, an effect of NC sentence type was only found on Chunk 6 ($b = 0.02$, $SE = 0.01$, $t = 2.79$, $p = 0.006$), as can be appreciated by looking at the model output in Table 2. The full output of the analyses can be found in our online repository (<https://osf.io/qv82h/>).

3.3. Discussion

The findings of our second experiment highlight a significant difference in the processing of inverse NCs compared to their canonical counterparts. Despite the two conditions sharing an identical surface form, the post-verbal Subject

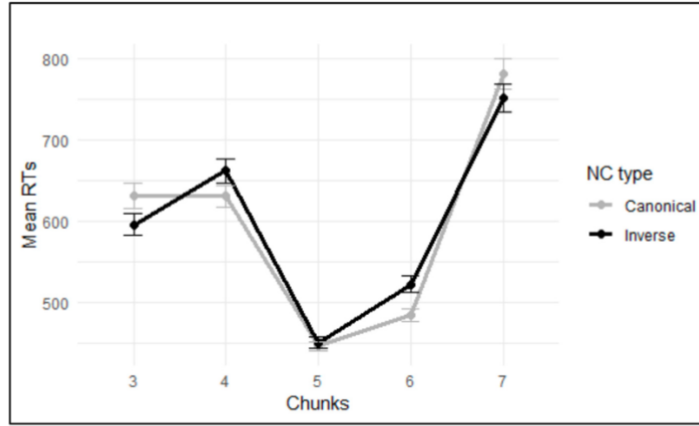


Fig. 2. Results of Experiment 2, with error bars representing standard error.

Table 2
Results of the linear mixed model fitted on Chunk 6: $\text{LogRT} \sim \text{Condition} + \text{Frequency} + \text{Length} + (1 | \text{ID})$.

Predictor	Estimate (β)	Std. Error	df	t	p	Significance
(Intercept)	6.15	0.03	78.99	244.44	<0.001	***
Condition inverse	-0.02	0.01	989.95	-2.88	0.004	**
Frequency	-0.01	0.01	991.04	-0.98	0.327	
Length	0.07	0.01	997.15	5.91	< 0.001	***

particularly entails higher cognitive effort. This is in line with [Moro's \(1997\)](#) raising theory arguing for a syntactic difference in inverse NCs. Additionally, these findings seem to contradict an underspecified syntactic representation of the sentence, even in the absence of specific probes afterwards. Since our analysis also considered word length and frequency, it is likely that differences in processing Chunk 6 are due to post-lexical features. However, one limitation of the self-paced reading paradigm is that readers must proceed strictly in a left-to-right sequence and cannot reread earlier parts of the sentence. This is a specific task effect, since during natural reading rereading is instead very common, especially when a specific part of the sentence is difficult to process. For this reason, the SPR methodology may be less effective at disentangling early from late stages of processing. Consistent with our research questions, to further delve into the analysis of role assignment in NCs we followed up this self-paced reading task with an eye-tracking while reading experiment.

4. EXPERIMENT 3

An eye-tracking experiment within the reading paradigm was carried out to examine more thoroughly the processing of canonical and inverse NCs. Experiment II seemed to point to difficulties associated with the post-verbal subject. Given the identical surface structures of the two conditions, this additional cognitive cost possibly stems from the assignment of syntactic roles. Therefore, a more fine-grained online methodology could help shed light on the syntactic integration occurring in inverse forms, revealed by late measures such as total reading times and regression path durations ([Clifton et al., 2007](#)).

4.1. Methods

Sixty-nine adult native Italian speakers (age range: 18–34; mean age: 25; 34 female, 26 male) participated in the eye-tracking experiment. Nine were excluded due to not being monolingual (3), having a diagnosis of dyslexia (2) or ADHD (1) or because they did not complete the experiment due to technical issues (3). All remaining 60 participants had normal or corrected-to-normal vision and no history of learning or language disorders. They all gave informed consent. The experiment was conducted using an SR Research Eyelink 1000. Reading was binocular, but data was collected

from the dominant eye at a sampling rate of 1000 Hz. Participants were seated approximately 60 cm from the screen, with a chin-rest to minimize head movements. The experiment included one practice block followed by two experimental blocks. Calibration was performed on a 9-point grid before the practice trials and repeated as necessary. Each trial began with a fixation point for drift correction, aligned with the start of the upcoming sentence on the left and centered vertically. Sentences were displayed in 24-point monospaced Courier New font, with black text on a light gray background. Participants could take as much time as they wished to read the sentence, and were instructed to press the spacebar once they were done reading to proceed to the next sentence. The experimental items were 14 in total and they were the same as those used in the two previous experiments, except for some modifications that were implemented to adjust the stimuli to the new methodology. Specifically, the sentences were modified with the aim of lengthening the copular verb, to minimise the chance of being skipped, as in previous studies (Paolazzi et al., 2022) this was considered a confounding element in the increase of reading times in the following regions. For this reason, the main clause of the experimental items was changed from “X said that” to “I believe that”, which in Italian then requires a subjunctive verb tense in the subordinate clause, realised as the copula “sia” in the case of our stimuli.

- (9) a. *Credo* *che* *la foto* *del* *muro* *sia* *la causa* *della* *rivolta.*
 Believe_{.1st.sing.pres} that the picture of.the wall be_{.3rd.sing.subj} the cause of.the riot.
 ‘I believe that the picture of the wall is the cause of the riot’
- b. *Credo* *che* *la causa* *della* *rivolta* *sia* *la foto* *del* *muro.*
 Believe_{.1st.sing.pres} that the cause of.the riot be_{.3rd.sing.subj} the picture of.the wall.
 ‘I believe that the cause of the riot is the picture of the wall’

Because this procedure can be more demanding for participants compared to self-paced reading paradigms, comprehension questions were included after only a quarter of the items, all of which were filler sentences. The same criterion for exclusion was established: participants who answered more than 25 % of these questions incorrectly would be excluded from data analysis. The task lasted approximately 20 min.

4.2. Data analysis and results

No participant was excluded from data analysis due to low accuracy on the comprehension questions. Two main interest areas were taken into consideration for data analysis:

Subject noun (word 4 in canonical and 9 in inverse NCs)
 Predicate noun (word 9 in canonical and 4 in inverse NCs)

For each interest area, we extracted First fixation duration, Gaze duration, Total reading time, Fixation count, Visit count and Regressions in. This analysis allowed us to detect potential effects of our experimental manipulation by comparing identical words, thus minimising potential confounding factors such as word frequency and length. Relevant reading measures were also collected for the copula and the last word of each sentence. Specifically, Regressions in and Skipping rate were examined for the copula, and Regressions out for the last word. We also extracted metrics traditionally associated with syntactic complexity (Clifton et al., 2007), namely Total reading time and Regression path duration, on the second head noun (word 9), regardless of its role as Subject or Predicate, considering its location within the critical sentence. As readers encounter each word in linear order from left to right, the second head noun, which follows the copula, constitutes the critical recognition point at which the non-canonical nature of the sentence is made evident. In establishing our strategy for data analysis, we opted for fitting models for Subject and Predicate nouns separately, then for word 9. This was done instead of analysing Subject and Predicate nouns together in a model including both sentence type and position as factors, thus

leading to a 2x2 factorial design. The choice was made starting from the assumption that a 2x2 analysis was not necessary for our purposes, but also taking into account two main reasons. First, it was meant to avoid excessive complexity in the fixed structure, as in addition to the two cited factors and their interaction, it would have been necessary to include the length and frequency of the nouns used. Secondly, an analysis including position as a factor would not have been adequate for all measures, for instance, not for Regression path duration, for which it is inaccurate to compare words that are at the beginning of the sentence to words that are at the end. For discrete variables such as Fixation count, Visit count and Regressions in, Skipping rate and Regressions out, generalised linear mixed models were fitted, assuming binomial or poisson distribution, using the `glmer()` function from the `lme4` package in R (Bates et al., 2015). The random structure of each model was established using the same data-driven approach described for experiment 2. As a preprocessing step for the reaction time (RT) data obtained by extracting First fixation duration, Gaze duration, Total reading time and Regression path duration, on the other hand, we adopted a model criticism approach following initial mild trimming (Baayen and Milin, 2010). At the first stage, RTs below 80 ms and above 1000 ms were excluded to remove anticipatory and excessively delayed responses. The remaining RTs were then log-transformed to approximate a normal distribution. Data analysis was then conducted using linear mixed-effects models (LMMs), accounting for both fixed and random effects, using the `lmer()` function (Bates et al., 2015). For the models analysing the Subject noun, Predicate noun, copula and last word, only condition (inverse or canonical) was included as a fixed factor in each model. For the models focusing on word 9, however, as two different nouns were being compared, the length and frequency of the words used was also added to the fixed structure of the models. The random structure was determined as above. Standardised residuals were subsequently extracted to assess model fit and identify potential outliers. Since none of the models returned more than 3.5 % of the residuals exceeding an absolute z-score of 2.5, no further data trimming was deemed necessary. We will now discuss the effects found during data analysis. As the data extracted from Gaze duration in the two main interest areas, as well as the measures taken for the copula and last word, did not reveal any effect, they will not be discussed any further. The full analyses can however be found in our online repository (<https://osf.io/qv82h/>).

First fixation duration. First fixation duration reflects the early stages of lexical access. In the case of the Subject noun, the model revealed a significant effect of condition, with longer durations in the inverse condition compared to the canonical one ($b = 0.09$, $SE = 0.02$, $t = 4.34$, $p < 0.001$), signalling that first fixation duration was longer for the subject when it was found in post-verbal position. No significant effect is found for the Predicate noun ($b = -0.03$, $SE = 0.02$, $t = 1.69$, $p = 0.092$), only a very marginal one, which suggests that it is not merely the position (pre-verbal or post-verbal) in the sentence that creates the effect for the Subject, but its syntactic contribution. The difference in durations and effect of condition for this measure can be appreciated by looking at Fig. 3.

Regressions in. This measure accounts for the number of regressive fixations (i.e., from right to left) made back on an interest area. For what regards regressions, there is a significant effect of inverse condition in the case of the Subject noun ($b = 0.29$, $SE = 0.13$, $t = 2.24$, $p = 0.025$), which shows that there were more regressions on the Subject noun when it was post-verbal (inverse condition) compared to when it was pre-verbal (canonical condition). No significant effect is found for the Predicate noun ($b = 0.07$, $SE = 0.10$, $t = 0.78$, $p = 0.455$). The fact that the effect of regressions is not replicated for both Subject and Predicate suggests that it is not reducible only to the noun being pre or post-verbal in reading order.

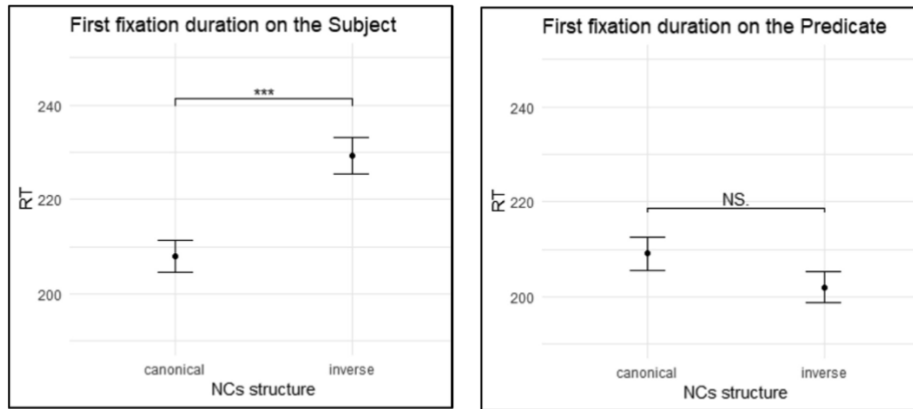


Fig. 3. Mean first fixation duration on the Subject and Predicate, with error bars representing standard error.

Fixation count and Visit count. These two measures generally correlate with other cognitive measures (e.g., total reading time) and count the times the eye visits an area of interest (Visit count) and the total number of fixations on that area (fixation count). In line with the results found for the number of regressions, in the case of the Subject noun there is a significant positive effect of the inverse condition for both measures ($b = 0.14$, $SE = 0.04$, $t = 3.29$, $p = 0.001$ for Fixation count; $b = 0.15$, $SE = 0.05$, $t = 3.13$, $p = 0.002$ for Visit count), while no significant effect can be detected on the Predicate noun ($b = 0.06$, $SE = 0.05$, $t = 1.39$, $p = 0.164$ for Fixation count; $b = 0.05$, $SE = 0.05$, $t = 0.97$, $p = 0.332$ for Visit count).

Total reading time. This measure accounts for the sum of all fixations made on an interest area, thus reflecting the cognitive effort required to process a word and integrating it in context. For the Subject noun, there is again a significant effect of condition ($b = 0.21$, $SE = 0.04$, $t = 5.56$, $p < 0.001$), which shows that people spend more time reading the Subject noun in the inverse structure, when it is post-verbal, than when it is pre-verbal. On the other hand, the Predicate noun is also read more quickly when it is pre-verbal, as shown by the significant negative effect of inverse condition in the model ($b = 0.12$, $SE = 0.05$, $t = 2.31$, $p = 0.025$). Therefore, the differences found for Total reading time could show an effect of sentence order: nouns are read more quickly when they precede the copula than when they follow it. The difference in reaction times and effect of condition for this measure can be appreciated by looking at Fig. 4. An analysis of Total reading time was also performed separately on word 9, which is the critical recognition point of the sentence (see above). Performing such analysis meant comparing the Subject and Predicate noun, which were realised with different lexical entries, in the post-verbal position. Thus, in these analyses, we added length in characters and frequency (extracted from *itTenTen20*, Jakubíček et al., 2013) as factors in the models.

The results, which can be appreciated by looking at Table 3, show that after the copula more time is used to process the Subject noun (616 ms on average) compared to the Predicate noun (516 ms on average) and although length is a significant predictor for this difference, as longer words were read more slowly ($b = 0.12$, $SE = 0.05$, $t = 2.64$, $p = 0.017$), the difference found not only reducible to that, given that condition also has a significant effect in the model ($b = 0.15$, $SE = 0.07$, $t = 2.22$, $p = 0.046$).

Regression path duration. The regression path duration constitutes the sum of the durations of all the fixations made from the first time the participant accessed word 9 until their eyes left the area, proceeding linearly towards the right. This means that this measure includes the duration of any fixation the participant has made going back to the left of the word for re-reading, before they proceeded to the next segment and it signals the processing effort involved by the sentence up to that point. The output of the linear mixed model can be seen in Table 4. In line with what was found for Total reading time, there is a significant positive effect of the inverse condition ($b = 0.16$, $SE = 0.05$, $t = 2.24$, $p = 0.004$). The length of the word presented at position 9 also had a positive effect on reading times ($b = 0.09$, $SE = 0.03$, $t = 3.35$, $p = 0.007$).

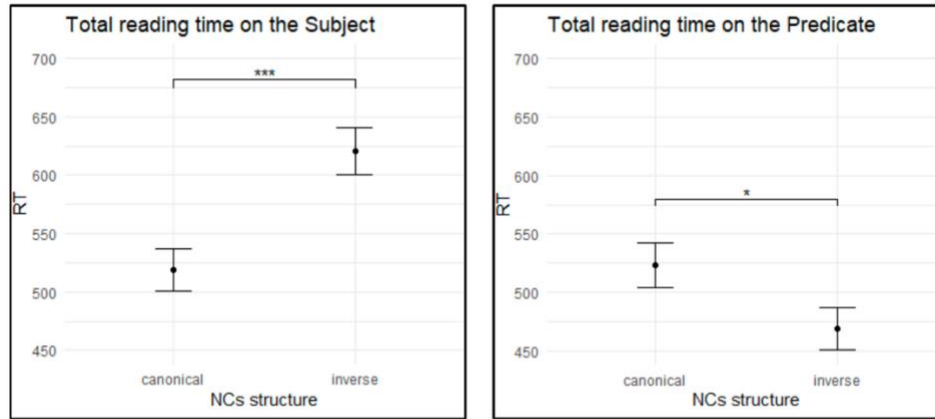


Fig. 4. Mean total reading time (RT) on the Subject and Predicate, with error bars representing standard error.

Table 3

Results of the linear mixed model fitted on Word 9 for total reading time: $\text{LogRT} \sim \text{Condition} + \text{Frequency} + \text{Length} + (1 + \text{Condition} | \text{Item}) + (1 | \text{ID})$.

Predictor	Estimate (β)	Std. Error	df	t	p	Significance
(Intercept)	6.07	0.05	46.79	111.55	<0.001	***
Condition inverse	0.15	0.07	12.25	2.22	0.046	*
Frequency	-0.08	0.04	14.62	-1.88	0.080	.
Length	0.12	0.05	17.57	2.64	0.017	*

Table 4

Results of the linear mixed model fitted on Word 9 for regression path duration: $\text{LogRT} \sim \text{Condition} + \text{Frequency} + \text{Length} + (1 + \text{Condition} | \text{Item}) + (1 | \text{ID})$.

Predictor	Estimate (β)	Std. Error	df	t	p	Significance
(Intercept)	5.59	0.05	20.70	106.44	<0.001	***
Condition inverse	0.16	0.07	10.87	2.24	0.005	*
freq	-0.01	0.03	11.02	-0.43	0.067	.
length	0.09	0.03	10.40	3.35	0.007	**

4.3. Discussion

The main findings of the eye-tracking experiment reflect the asymmetry of NCs and support the hypothesis that syntactic derivation alone can be the cause of additional processing cost during reading.

A significant effect of the NCs type, inverse or canonical, is found mainly on the subject. Sentential subjects are looked at more often and for longer times in inverse copular sentences, when they appear in post-verbal position. This effect is seen through a higher proportion of regressions in, visit count and fixation count, as well as longer total reading times and first fixation durations. An effect of sentence type is instead not as clearly seen for the sentential predicate, for which the only significant effect detected regards total reading time. The results for this measure show that noun phrases generally require longer reading times when they are found in post-verbal position, regardless of whether they fulfil the role of subject or predicate in the sentence. Additional analyses, however, underlined a distinction in reading patterns on the post-verbal noun head, shown through both total reading time and regression path duration, which do not seem to be reducible to sentence order, word length or frequency. The findings obtained by analysing Total reading time and Regression path duration on the sentence's recognition point (word 9) are of particular interest, because they reflect increased processing cost for inverse NCs. These two measures are traditionally associated with syntactic complexity, further supporting the raising analysis. Overall, these findings show that sentence order affects reading times, with nouns in post-verbal positions requiring more processing time. Regardless of this effect, however, a specific effect of syntactic derivation is also found.

5. GENERAL DISCUSSION

The present study aimed at exploring the processing of canonical and inverse nominal copular sentences. In spite of their presenting a stark contrast between identical linear surface structures and deep syntactic asymmetries, these sentences still represent an understudied topic in experimental syntax. We will discuss the results from our three experiments in light of current theoretical and parsing models and also highlight the contributions of various methodologies to this study along with its limitations.

5.1. Main findings

Experiment 1 used an acceptability judgment task, focusing on participants' conscious evaluations of sentence naturalness and comprehensibility. In an attempt to further investigate Greco et al.'s (2020) findings on inverse copular sentences, the experiment used the same procedure (a 7-point Likert scale) and an adapted version of the same materials while doubling the participants and keeping the copular type as the only factor. Greco et al.'s (2020) original sentences leveraged one distinctive trait of Italian nominal copular sentences, that is the copula agreement and used plural subjects to assess the acceptability of canonical and inverse structures. They found inverse NCs to be perceived as "barely grammatical", with an average rating below 4.5, arguing for a violation of the subject position. In a further acceptability judgment task on unergative and unaccusative constructions, the authors also confirmed a general preference for pre-verbal subjects among adult Italian speakers. While being in line with theoretical expectations about Italian, an SVO language, Greco et al.'s (2020)'s results do not speak about the several linguistic and cognitive factors of sentence processing that are thought to be linked to behavioural responses in acceptability judgements (Sprouse, 2023). As both sentences are grammatical and straightforward violations are not at issue, expectations contrary to a subject-first strategy could weigh disproportionately in processing and later judging these specific sentences. If this was the case, avoiding the overt cue of the plural agreement would mitigate the difference in acceptability between the two levels under investigation. This prediction was confirmed by our results, since participants judged both canonical and inverse NCs as equally natural and comprehensible when morphosyntactic cues such as number mismatches were eliminated. It is still worth noting, however, that the divergence did not reverse the outcome, as the canonical copular sentences scored roughly the same between the two experiments, and always above the inverse sentences.

Our results from Experiment II showed that online processing of inverse NCs is indeed more costly compared to canonical NCs. Participants display significantly slower reaction times on the post-verbal subject, thus confirming a possible preference for pre-verbal subjects already observed in Authors' et al (2020). However, in the absence of overt cues signalling the subject-verb agreement, this preference is only detectable online. Since the processing of NCs has not been investigated in previous research, we discuss these results with regard to two main points raised by our study. Firstly, when non-canonical forms, such as "inverse" NCs, present the same exact surface structure of their canonical counterparts i.e. DP-copula-DP meaning that the reading times of incoming materials cannot be explained by build-up expectations, they are indeed more difficult to process. Secondly, even in the absence of a syntactic probe, i.e., a question specifically targeted at the syntactic roles within the sentence right afterward, our participants seem to correctly assign syntactic roles to the two NPs, as evidenced by the differential processing cost associated with the post-verbal Subject compared to the post-verbal Predicate.

Finally, Experiment 3 confirms and expands on our previous findings, showing significant differences in the online processing of inverse copular sentences as opposed to canonical copular sentences. Moreover, this effect was found specifically in late measures, traditionally associated with sentence processing difficulties and later reanalyses. Longer total reading times and regression-path durations, more regressions and a higher visit count were found on the post-verbal subject, with post-hoc analyses excluding any position effect within the sentence. That means that

participants tended to reassess their initial representation once they encountered the post-verbal NP, only in the case this was the subject. Therefore, this finding corroborates the preference for pre-verbal subjects in Italian, already observed in [Greco et al.'s \(2020\)](#), by showing that the parser initially assigns syntactic roles based on an SVO order.

5.2. Theoretical frameworks

The present study investigates Nominal copular sentences in Italian. In doing so, we adopt [Moro's \(1997\)](#) interpretation of the structural asymmetries observed between canonical and inverse NCs, namely the raising theory. He postulates that two DPs are merged into a small clause, leading to an unstable structure from which either the DP Subject or the DP Predicate needs to raise in the pre-verbal position. In the former case, the outcome would be a canonical nominal copular sentence, whereas in the latter this movement results in an inverse nominal copular sentence. The latter constitute a very peculiar case of non-canonical structure, in which the reversed word order gives rise to an identical surface string of elements compared to its canonical counterparts, i.e. DP V DP. Although theoretical accounts do not always allow for clear psycholinguistic predictions, as they are not intended to explain cognitive behaviour, the raising theory establishes the inverse NC as a more complex structure with respect to the canonical NCs, given the different subadjacency conditions between the two NPs. In observing the processing of NCs, therefore, we expect to find a more effortful reading in inverse NCs. This prediction is indeed confirmed by our results, which show increased reading times in late-stage processing, probably due to reanalysis of the roles initially assigned to the two DPs. Taken together, the findings from our three experiments are also consistent with a general subject-first preference which, in the absence of morphosyntactic cues to verbal agreement, is not picked up consciously by native speakers, but emerges rather unconsciously during online processing. In the specific case of nominal copular sentences, therefore, we found that post-verbal subjects are cognitively costly. We argue that our results highlight nonetheless one further asymmetry between subject and predicate in NCs, calling for further experimental assessment of the current theoretical frameworks that are able to exhaustively account for the differences between canonical and inverse NCs.

Finally, we acknowledge the possibility that increased processing costs might be also related to the relative frequency or predictability of inverse copular constructions, as suggested by an anonymous reviewer. While we do not rule out potential effects of frequency or structural expectancy, our interpretation emphasizes syntactic derivation as a theoretically grounded factor contributing to processing difficulty: if frequency were the key factor in accounting for the results shown in this paper, we would expect to find similar differences in the acceptability judgment task. In fact, this is not the case, further suggesting that frequency alone is not sufficient to explain the observed effects. Moreover, to our knowledge, there is no available corpus data that systematically compares the frequency of inverse vs. canonical copular sentences in Italian. Future studies will shed light on this point. Finally, inverse structures are not inherently rare – existential constructions (see [Moro, 1997](#)) are frequent and widely analyzed as inverse configurations.

5.3. Task differences

As observed in discussing the main findings, our Experiment I did not confirm previous results on the same task, that is acceptability judgments on canonical and inverse NCs by adult native speakers. We suggested that a subject-first strategy may lead to a lesser perceived naturalness in the presence of a number mismatch between the two DPs and a post-verbal subject. The question arises as to whether expectation can fully account for the differences observed in [Greco et al.'s \(2020\)](#) study or, alternatively, syntactic derivation still plays a role in sentence processing that is not revealed by offline measures. This is not to say that acceptability judgements are not sensitive to most kinds of syntactic effects, as decades of experimental literature would otherwise prove, but

rather than the asymmetries under examination may be too subtle to be consciously picked up in a metalinguistic judgment task (a similar issue is debated in [Linzen and Oseki, 2018](#)). Such an approach would agree with [Sprouse's \(2008\)](#) take on the reanalysis confound, according to which processing difficulty without ungrammaticality comes with no judgment cost. From this perspective, the processing effects stemming from the subject-predicate asymmetry in NCs would arise from a temporary representation (arguably, a subject-first structure) which is then reanalysed and reassessed.

To explore this possibility, we discuss here the results with respect to online methodologies as well. Experiment 2 employed a self-paced reading paradigm to investigate sentence processing. Self-paced reading has been extensively employed in experimental studies on syntax, as reaction times are considered to reflect the cognitive load assigned to each region of interest. More specifically, it has proven a successful tool in sentence processing, for its sensitivity to ambiguities and garden-path structures, which are particularly relevant to our study in terms of reanalysis processes while reassigning thematic roles. In line with predictions based on a deep structural (re)analysis, we found increased reaction times on the post-verbal subject in inverse NCs. We find this result particularly interesting in light of the comparison to Experiment I, which highlights the difference between conscious linguistic judgment and unconscious linguistic comprehension. Confirming the reanalysis confound suggested for Experiment I, evidence of a cognitive load associated with syntactic derivations is detected in reaction times during online processing. Since the masked self-paced reading paradigm induces a strictly sequential reading, however, we decided to employ a more fine-grained methodology that would take into account late measures and re-readings as well. Self-paced reading measures are often compared to a first-pass reading when, in reality, they might conflate early and late stages of processing, depending on the reader's way of dealing with the impossibility of revisiting previous segments. Relevant to our study, evidence for late-stage difficulties emerging from reanalysis come, among others, from studies on non-canonical structures and ambiguity in reference resolution. In an eye-tracking experiment following up a SPR task, for example, [Paolazzi et al. \(2022\)](#) found an interaction between event structure and passivisation on late measures only. In another study on reflexive anaphors, this time involving eye-tracking, [Sturt \(2003\)](#) observed that binding violations (i.e., a gender mismatch with the accessible antecedent) caused immediate disruption in early measures, while consequently impacting late measures when the inaccessible antecedent matched the reflexive pronoun. The authors interpreted this as a recovery strategy, forcing the parser to reconsider ungrammatical interpretations of the anaphoric reference in the absence of a valid licit antecedent. In our study, Experiment III also found significant differences in later measures, particularly total reading times and regression path duration, possibly signalling the reassessment of syntactic roles assigned during first-pass readings. This goes to show that while the asymmetries in NCs are evidently syntax-driven, the processing effects associated with their resolution are presumably due to reanalysis, which is hardly captured by experimental paradigms that do not rely on time-locked observations. This calls for the integration of a variety of paradigms when questioning specific processing mechanisms, so that syntactic theory can stand on solid empirical grounds.

Finally, our design differs from other studies on non-canonical sentences in two main aspects: first, it does not test (mis)interpretations, also because our stimuli do not leave room for semantic ambiguity as was the case for previous research ([Ferreira, 2003](#); [Cutter et al., 2022](#)); second, it does not require a secondary task specifically probing the syntactic roles of the sentences. Although we cannot directly compare the task effect induced by questions about the superficial or the detailed meaning of the sentence, we agree with [Bader and Meng \(2018\)](#) that task demands may induce a retrieval confound in testing the interpretation of non-canonical sentences, making difficult to generalize those results to the representations associated with those sentences. Although further research is needed, results from the three paradigms employed in the present study also point to a discrepancy between implicit and explicit analyses during sentence comprehension.

5.4. Human parsing mechanisms

Our analysis focuses on the acceptability and the online processing of canonical and inverse nominal copular sentences, which differ in their deep structure but not in their linear surface order. This contrast, paired with reading for comprehension paradigms, gave us insights into the parsing mechanisms involved in sentence processing. First and foremost, Experiment II and III consistently showed that the parser has access to the syntactic structure of the sentence, being able to process differently canonical and inverse NCs. This difference, already detected through higher reaction times on the post-verbal subject in a self-paced reading task (Experiment II), is further confirmed in the eye-tracking while reading experiment. The parser seemingly undertakes a reanalysis of the syntactic roles previously assigned, as evidenced by the inflated reading times in late measures such as total reading time and regression path duration. Consistently with the SPR, this process of re-evaluation begins fairly early into the sentence, as indicated by the significant difference in the regression-path duration already on the post-verbal subject, that is arguably the recognition point for the algorithmic parser to step in. Our data seem to argue against an underspecified syntactic representation occurring during language comprehension. Given the absence of specific probes required within the task, we found evidence of algorithmic processes operating regardless of the task at hand and correctly interpreting the NP syntactic roles and their position with respect to the copula.

Finally, it is worth noting that our experiments were designed to isolate and control for certain cues, specifically underlying syntactic cues. Comparing our results from Experiment I to what had been previously found by [Greco et al.'s \(2020\)](#) already highlighted the interaction of multiple cues in judging the acceptability of the NCs. Therefore, by eliminating the surface morphosyntactic cues to verb agreement, we were able to better understand the emergence of algorithmic mechanisms. It is possible, and expected, that changes in additional sources the parser rely on, such as prosody or overt cues to syntactic and semantic roles, lead to even stronger effects in inverse NCs, maybe detectable already in first-pass readings.

5.5. Limitations and future research

Finally, some limitations should be acknowledged. First, the study primarily focused on syntactic derivation as the driving factor behind the processing asymmetries between canonical and inverse NCs. However, additional factors, such as prosodic features, semantic plausibility, or pragmatic inferences, could interact with syntactic structures to influence comprehension. Future studies could systematically manipulate these variables to assess their role in shaping NC processing. In the same line, although differences between canonical and inverse NCs persist across languages, the experimental investigation of these structures in other languages could highlight the differential weight of multiple cues in online parsing. For example, [MacWhinney and colleagues \(1984\)](#) found that Italian native speakers are mostly guided by agreement patterns to predict the identity of subject and object, arguing that although word order in Italian is an applicable cue, it is not as reliable. Therefore, according to the Competition Model “the most valid cue to subject assignment in Italian is agreement marking on the verb” ([MacWhinney et al., 1984:147](#)), at least to sentence interpretation. This is in line with our first experiment contradicting what was previously found by [Greco et al.'s \(2020\)](#) once deleted the overt cue to verb agreement. However, we also found increased processing in interpreting inverse NCs during online processing, challenging models that heavily rely on surface cues, such as the Competition Model ([MacWhinney et al., 1984](#)) as well as the extended Argument Dependency Model ([Bornkessel-Schlesewsky and Schlewsky, 2016](#)). Nonetheless, looking into diverse languages could help further investigate the time-course of the integration of possible surface cues as well as their validity in specific contexts- and syntactic movements into the processing of canonical and inverse forms, perhaps revealing an impact on early measures as well.

6. CONCLUSIONS

The results of our experiments provide support for the raising analysis of nominal copular sentences (NCs), as proposed by Moro (1997, 2017), by showing that inverse NCs incur greater processing costs compared to their canonical counterparts, especially when it comes to the post-verbal subject. The increased processing difficulty for inverse NCs likely reflects the structural cost required for predicate DP raising, as opposed to the subject DP raising in canonical NCs. This distinction highlights the role of syntactic dependencies in defining asymmetries in sentence processing. The absence of confounding factors, such as number mismatch between subject and predicate, strengthens the conclusion that the observed effects stem from the inherent syntactic derivation of inverse NCs.

Moreover, the findings challenge purely heuristic models of sentence processing, which would predict no differences given the identical DP-is-DP order of both NC types. Similarly, our results suggest that prediction cannot entirely account for the processing effort required by the sentence, and that this is better highlighted by integrating different experimental paradigms with varying degrees of sensitivity. Instead, these results support hybrid models, where comprehension goes through a dual-route process, involving both a heuristic processing and a deeper algorithmic procedure. In our study, participants seemingly start from a subject-first representation, consistent with the SVO structure of the language under examination and then reanalyse the sentence according to the copular type. More specifically, reanalysis occurs only in the case of inverse copular sentences, i.e., when the first heuristic representation is not confirmed by the deep parser. Crucially, this differential behaviour among conditions with equal tasks suggests that a fully specified syntactic representation is always achieved. Therefore, although misinterpretation effects are beyond the scope of this study, our results seem to exclude persistent inaccurate or “good-enough” representations. The contrast between canonical and inverse NCs under Author 4’s raising analysis demonstrates that syntactic relations drive processing difficulties over surface linear order and helps to shed light on how human parsing mechanisms navigate complex sentence structures.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Veronica D’Alesio: Writing – review & editing, Writing – original draft, Methodology, Investigation. Anna Teresa Porrini: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. Matteo Greco: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. Andrea Moro: Conceptualization.

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Data availability

We have shared the (anonymized) link to our data in our manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX A. SUPPLEMENTARY MATERIAL

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lingua.2025.104019>.

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